Can LLMs Detect Errors in Long Chain-of-Thought Reasoning?

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Motivation and Problem

Have you ever given the right answer by pure luck — even though your method was totally wrong?

- Users *trust* LLMs.
- Errors in reasoning => real-world consequences (healthcare, coding, education).
- Trusts requires self-checking e.g pilots => copilots

Paper Contributions

This paper asks a new and important question:

Can we train LLMs to catch mistakes in their own chain of reasoning?

✅ **1. Introduced a novel task:**
"Detect reasoning errors in long CoT (Chain-of-Thought)."

✅ **2. Built a new dataset:**
Annotated ~30K samples with correct and incorrect CoTs.

✅ **3. Proposed two approaches:**

- **Self-Evaluation:** LLM critiques its own CoT.
- **Verifier Model:** Separate model to evaluate CoT correctness.

✅ **4. Benchmarked performance:**
Compared with humans and baselines.

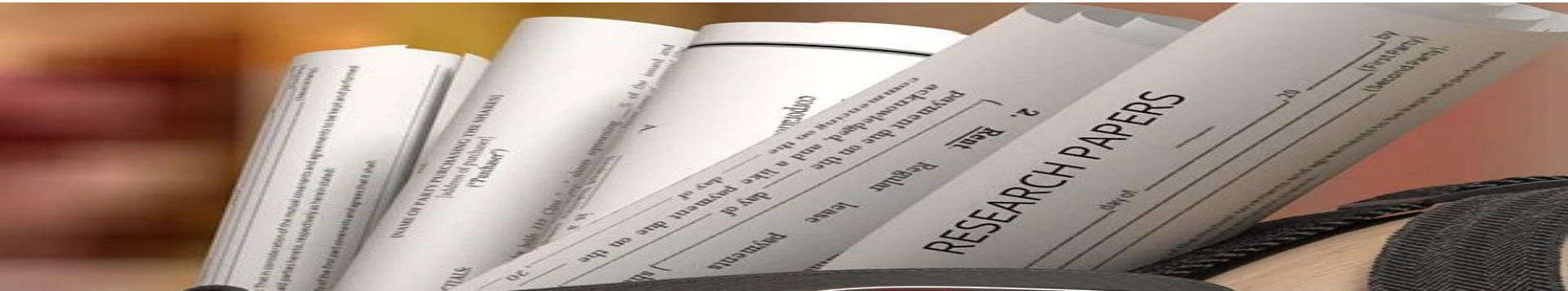
Prior Work vs This Paper

Chain of Thought (Wei et al., 2022): Prompt models to "think step-by-step" → solve harder problems better.

Self-Consistency (Wang et al., 2022): Generate multiple reasoning paths → pick majority answer.

Reflexion (Shinn et al., 2023): Models "reflect" on mistakes during multi-turn tasks.

BUT: Focused mainly on getting the *final answer* right, not checking if *reasoning steps* made sense!



What's Truly New

Innovation:

- Focus on **reasoning quality**, not just outputs.
- Open domain, diverse problems.
- Trainable across **different reasoning tasks**.

Aspect	Prior Work	This Paper
What was verified?	Final answers	Reasoning chain (CoT)
Dataset size	Small / Narrow	30K diverse examples
Methods	Classifiers, heuristics	Self-reflective + Verifier

Step By Step Process

- **Dataset Construction**: Generate diverse long CoTs using multiple models.
- **Segmentation**: Break CoTs into sections for easier checking.
- **Query Selection**: Pick diverse questions across domains (math, science, coding).
- **Difficulty Balancing**: Ensure mix of easy, medium, hard questions.
- **Data Cleaning**: Filter low-quality samples with GPT-4 and rules.

- **Human Annotation**: Experts check sections for correctness, usefulness, reflection.
- **Performance Metrics**: Use precision, recall, F1-score for evaluation.
- **Error Analysis**: Categorize frequent error types (e.g., calculation, logic).
- **Critique Testing**: Compare critic models' abilities to spot mistakes.
- **Insights**: Recommend ways to build better self-checking AIs.

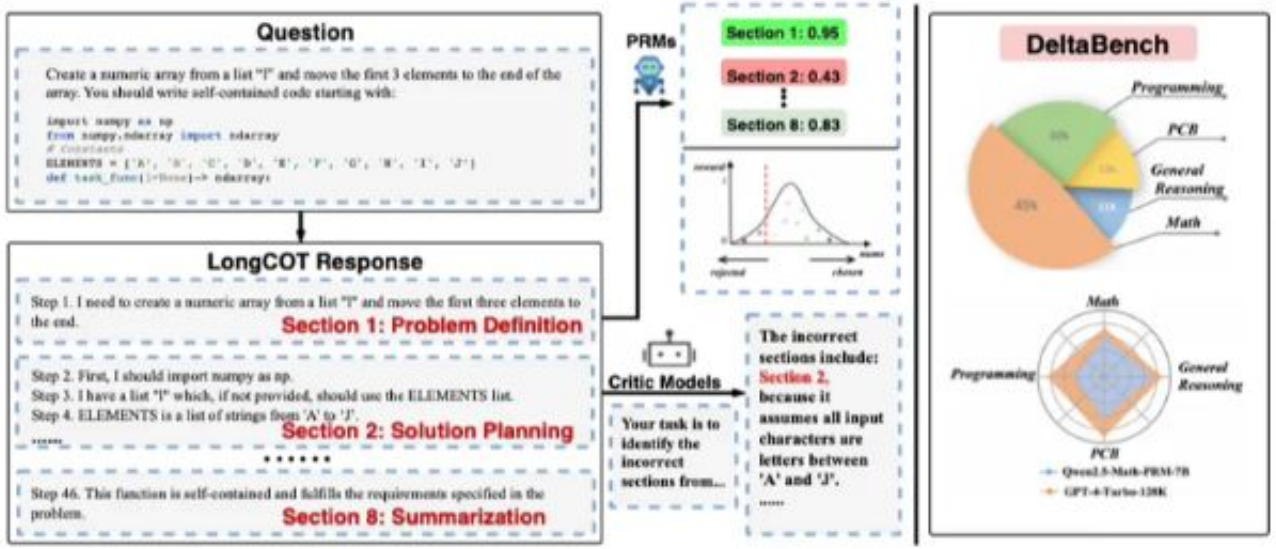


Figure 1: Illustration of the evaluation process for critic models and Process Reward Models (PRMs) for DeltaBench.

The Challenge

Chain-of-Thought = A Journey

E.g Question: 'If you have 5 apples, and you give away 3, how many are left?'

AI Reasoning:
5 + 3 = 8 → 8 apples left.

Wrong! But it *sounds* confident.

- Long chains = more places to slip.
- Mistakes can be hidden deep inside.
- Surface correctness ≠ Logical correctness.

Key Concepts & Goal

Key Concepts:

- **o1-like Models:** Generate detailed, step-by-step reasoning.
- **Critique Abilities:** Ability to detect errors within reasoning steps.

Research Goal:

- Introduce **DeltaBench** to evaluate how well models can find reasoning errors.

DeltaBench Dataset

Purpose:

- Analyze the effectiveness of long CoT reasoning generated by o1-like models.
- Measure how well models can critique these reasoning chains.

Models Involved:

- QwQ
- O-1
- DeepSeek-R1
- Gemini 2.0 Flash Thinking

Reasoning Tasks:

- Math
- Code
- General Reasoning

Evaluation Focus:

- Accuracy and usefulness of long CoTs.
- Ability to detect errors and reflections.
- Critique abilities of Process Reward Models (PRMs).

Comparison to Other Benchmarks

Key differences:

- DeltaBench focuses on difficult, long CoT problems
- Section-level evaluation, rather than sample-level or step-level
- Emphasis on detecting errors throughout the reasoning process rather than just the first error

Fine-Grained Analysis of Long CoTs

Common Problems in Long Reasoning Chains

- **Fundamental Errors**: Frequent mistakes in calculation, syntax, and formatting.
  → 25% error rate in QwQ-32B, 23% in Gemini 2.0.

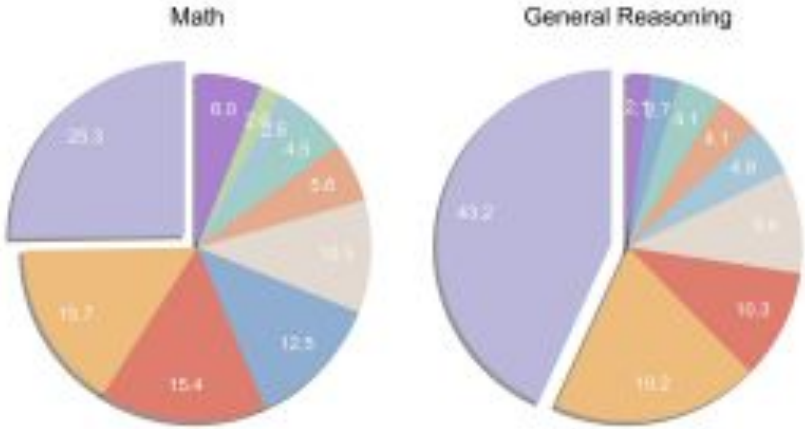
- **Reflection Ineffectiveness**:
  → Only 32.2% of model reflections lead to correct revisions.

- **Reasoning Redundancy**:
  → 27% of CoT content is unnecessary or repetitive.

Analysis Overview

Most Common Errors Across Domains

- Reasoning Error
- Understanding Error
- Calculation Error
- Strategy Error
- Formal Error
- Premise Error
- Knowledge Error
- Programming Error



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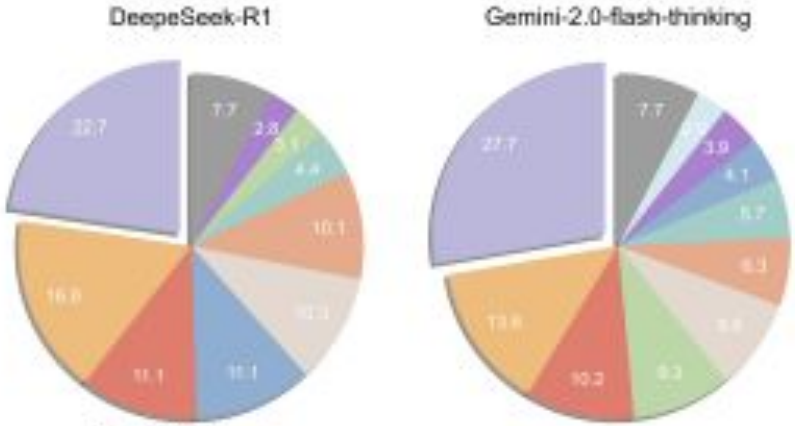


Main Results and Trends from DeltaBench

PRMs:

- Qwen2.5, Llama3.1, Skywork-o1, and others.
- Low accuracy on reasoning error.
- Performance does not improve with larger models.
- PRMs show more stability across CoT lengths.

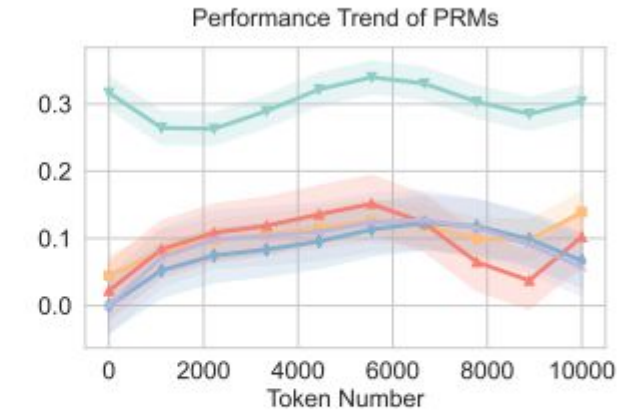
Critic Models:

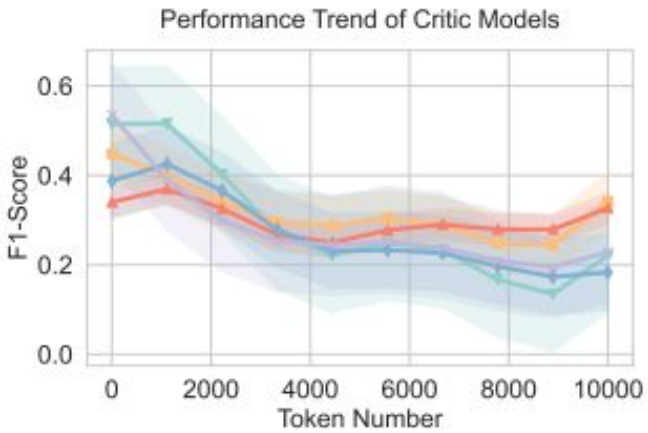
- Critic models include GPT-4, Gemini, Qwen models.
- GPT-4-turbo-128k outperforms others.
- Critic models' performance drops with longer CoT chains.

Effect of Long CoT Length

- **Critic models:** Performance drops with increasing token length.
- **PRMs:** Stable performance across token lengths.

Key Finding: Critic models struggle with long contexts, but PRMs show stability.





Evaluation of Critique Abilities

Critique Performance:

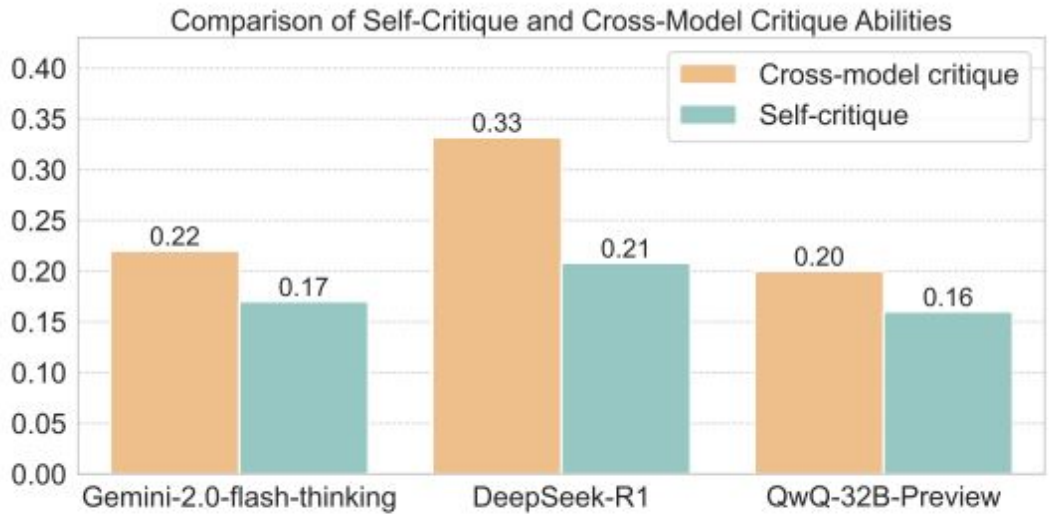
- Best model: GPT-4-turbo-128k (F1 score = 40.8%).
- Existing LLMs and PRMs still struggle with detecting reasoning errors reliably.

Comparison Across Models:

- o1-like models show no major advantage over non-o1 models in critique ability.
- Self-Critique Challenges:
- o1-like models perform worse at critiquing themselves than critiquing others.
- Example: DeepSeek-R1 shows a 36% drop in self-critique performance.

Reflection and Reasoning Quality

- DeepSeek-R1 had the highest reflection success rate.
- Reflection is more effective in math domain.
- Overall, valid reflection rates remain low.
- Average reasoning usefulness: 73% useful, 27% redundant.



Related Works

Test-time Scaling Methods:

- Tree search techniques to improve reasoning.
- Reinforcement learning (RL) methods to generate longer, better CoT outputs.

Process Reward Models (PRMs):

- Focus on rewarding **correct reasoning steps** instead of only final answers.
- Growth of **human-annotated** process-level datasets for training.

LLM Critique Capabilities:

- LLMs generate critiques and verdicts to spot errors.
- **Self-critique** methods help models improve their own reasoning.

CONCLUSION

DeltaBench: A comprehensive evaluation benchmark to analyze model ability to detect reasoning errors.

Key Contributions:

- Provided error analysis of models.
- Detailed comparison between **critic models** and **PRMs**.
- Highlighted model performance

Limitations:

- High cost of dataset annotation.
- Hard to scale across all domains due to domain-specific variations.

Areas for Future Work:

- Improve self-critique ability.
- Expand dataset coverage.
- Reduce annotation costs.

Thank You!

Any Questions?